Fiscal Year:	FY 2013	**Task Last Updated:**	FY 12/24/2013
PI Name:	Kennedy, Ann R. D.Sc.		
Project Title:	Center of Acute Radiation Research		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-02 Center of Acute Radiation Research (CARR)
Start Date:	09/01/2008	**End Date:**	04/30/2014
No. of Post Docs:	10	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2014 per December 2013 NSBRI report submission; Acute03 gap deleted per IRP Rev E (Ed., 1/8/14) Note: title change per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Freund, Gregory (University of Illinois at Urbana-Champaign) Wan, X. (University of Pennsylvania) Weissman, Drew (University of Pennsylvania) Cengel, Keith M.D., Ph.D. (University of Pennsylvania) Ebert, Douglas (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-RE01801		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	In this research, animal model systems are being utilized to evaluate the space radiation induced acute radiation syndrome (ARS) with vomiting (and/or retching) and white blood cell counts in ferrets; white blood cell counts, fatigue, and immune system parameters in mice; and skin injury, with accompanying immune system changes, and white blood cell counts in pigs as the biological endpoints. In addition, the effects of combined exposure to simulated hypogravity and space radiation on blood cell counts and immune functions with respect to both the innate immune system and the acquired immune system in mice are being evaluated in mice. The first few years of the project are focused on determinations of whether certain parts of the ARS occur in animals exposed to space radiation(s) simulating that occurring during a solar particle event (SPE). If problems are observed to exist for the model systems used, later years of the project will be focused on the development of countermeasures for these effects. In these studies, selected drugs will be evaluated in the animal model systems to determine their effectiveness in prevention and/or mitigation of the specific ARS symptoms. The dose response relationships measured for these endpoints will be compared mathematically with published results for human subjects to determine the RBEs and the predictive value of these animal models for bio-dosimetry and for intervention studies to evaluate FDA-approved drugs that can be used for prevention and/or mitigation of the ARS symptoms. Various tissues will be collected from the animals for histopathological examination and for studies by various immunohistochemistry, qPCR, and/or immunochemistry techniques to study the underlying mechanisms for the acute radiation syndrome and its prevention and/or treatment by drugs. As part of the studies in year 3 of this grant's activities, countermeasure studies began to identify agents that could prevent and/or mitigate the effects of SPE radiation on ferret emesis and mouse blood cell counts. These investigations were designed to address Acute Radiation Syndromes - Specific Gaps 1, 2, 3, 5, 6, 7, and 8 identified in NSBRI-RFA-08-02.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	It is well-known that exposure to ionizing radiation at sufficiently high doses can result in various types of adverse biological effects. The countermeasures to be developed as part of this research program have significance for all individuals exposed to high doses of ionizing radiation on Earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced adverse biological effects on Earth. The recent findings from CARR studies that doses near 2 Gy cause bacterial translocation and result in a systemic immune response may have great significance for clinical radiation therapy patients on Earth. It has been known for some time that considerably higher radiation doses (>2 Gy) cause bacterial translocation (i.e., those doses associated with the gastrointestinal (GI) tract syndrome), but it has not been considered likely that doses in the range of 2 Gy would have such effects. It is our hypothesis that much of the sickness experienced by radiation oncology patients, who are often exposed to 2 Gy doses of radiation, could be due to circulating bacterial products (such as lipopolysaccharide {LPS}) resulting from the radiotherapy treatments; thus, antibiotics could be helpful in reducing the nausea and other symptoms in patients during their radiation therapy treatment periods.		

Task Progress:	A study utilizing aged and young mice was performed to determine whether the RBE values previously determined in mice aged 6-8 weeks old were comparable to the RBE values determined in mice that were 1 year of age. Results: the age of the mice did not influence the response of white blood cells to total body SPE-like proton irradiation. Similar studies were performed to determine whether there were gender differences in the responses of mice to radiation exposure. Exposure to 0, 0.5, 1, and 2 Gy total body irradiation resulted in similar effects on white blood cells, lymphocytes, and granulocytes in male and female C3H/HeN mice, while red blood cell counts and hematocrit values remained stable post-irradiation. Non-irradiated male mice had 13% higher platelet counts, compared with their female counterparts, and showed enhanced recovery of platelets at day 16 post-irradiation. Thus, gender differences may influence the response of platelets to total body irradiation. It has been observed that a non-toxic bacterial challenge (with either Pseudomonas aeruginosa or Klebsiella pneumonia) leads to a high level of mortality in mice exposed to a dose of 2 Gy gamma or SPE-like proton radiation both with and without additional exposure to simulated microgravity (hindlimb suspension). The analysis of lower doses of gamma radiation with hindlimb suspension in bacterial challenge studies using C3H/HeN mice and Pseudomonas bacteria demonstrated that 1.5 Gy gave similar levels of morbidity as 2 Gy (90%), but 1.0 Gy of radiation resulted in only 30% morbidity, which is similar to hindlimb suspension alone. G-CSF (Neulasta) used as a countermeasure for bacterial challenge reduced morbidity from 80-90% to 20-30% with both gamma and proton radiation and hindlimb suspension in C3H/HeN mice challenged with Pseudomonas bacteria. Pigs exposed to skin doses of 2.5 Gy - 7.5 Gy simulated SPE - electron radiation (October 1972 event with deeper dose distribution) or a skin dose of 4.5 Gy from simulated SPE-proton radiation, had decreased WBC counts that did not return to baseline values by 60 days post-radiation. A 2 Gy dose of SPE-like proton radiation in pigs resulted in significantly decreased WBC and neutrophil counts; in pigs exposed to the same radiation conditions and treated with Neulasta, the WBC and neutrophil counts remained at baseline levels throughout the experimental period. Neulasta is an effective countermeasure for circulating blood cell reductions resulting from exposure to SPE-like radiation in pigs. Irradiated ferrets and pigs developed disseminated intravascular coagulation (DIC) in these studies. The results of experiments performed as part of this project, along with data from the published literature, have led to the hypothesis that the development of DIC is a major cause of death in mammals resulting from exposure to relatively low doses of radiation.
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